

PaLM-E: An Embodied Multimodal Language Model

Danny Driess, Fei Xia, Mehdi S. M. Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzaan Wahid, Jonathan Tompson, Quan Vuong, Tianhe Yu, Wenlong Huang, Yevgen Chebotar, Pierre Sermanet, Daniel Duckworth, Sergey Levine, Vincent Vanhoucke, Karol Hausman, Marc Toussaint, Klaus Greff, Andy Zeng, Igor Mordatch, Pete Florence

Embodied Language Models

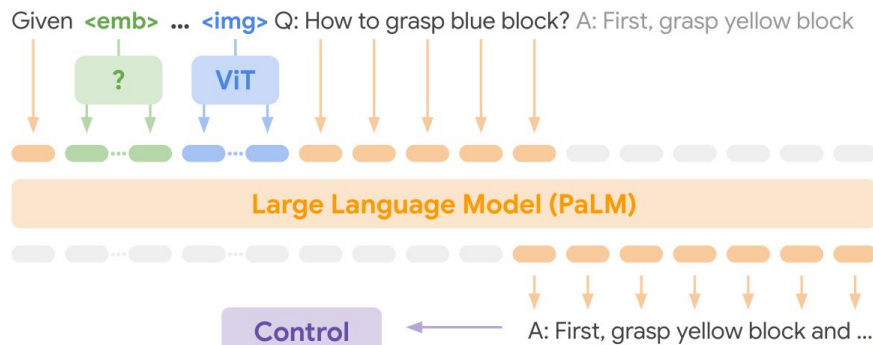
- incorporate real-world **sensor and actor modalities** into **language models**
- establish the link between words and percepts
- Input: multi-modal sentences
 - **images**
 - **text**
- Output: multi-modal sentences
 - **text**
 - **motor commands**
- trained end-to-end
- Model: pre-trained LLM
- Applications
 - **robotic manipulation planning**
 - visual question answering
 - captioning

Driess, D., Xia, F., Sajjadi, M. S., Lynch, C., Chowdhery, A., Ichter, B., ... & Florence, P. (2023). Palm-E: An embodied multimodal language model. arXiv preprint arXiv:2303.03378.

"One model"

- Embodied robotics tasks
- Vision-language
- Language
- ... across multiple robot embodiments
- ... across multiple modalities (vision, states, neural scenes)

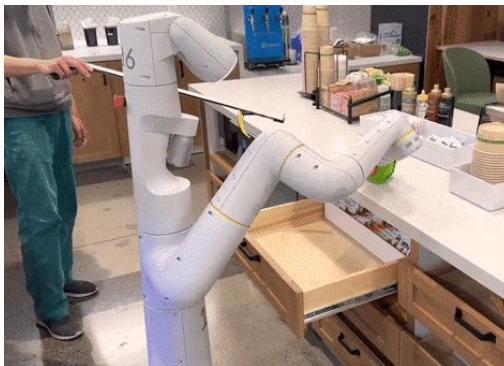
PaLM-E: An Embodied Multimodal Language Model



positive transfer

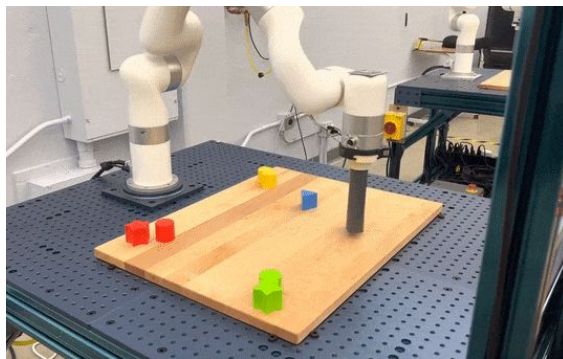
closed-loop end-to-end planning

("Given ****... Bring me the rice chips from the drawer")



long-horizon tasks

("Given ****.... Sort the blocks by colors into corners")



vision-language generalist



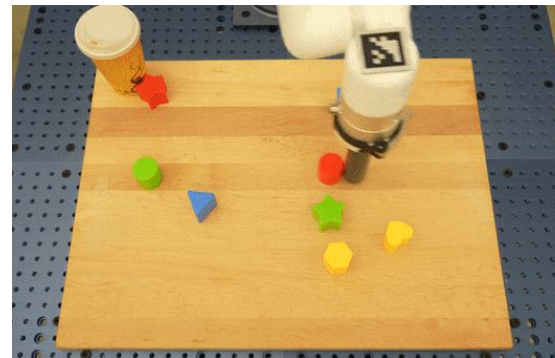
Given ****. Q: What's in the image? Answer in emojis.
A: 🍏 🍊 🍇 🍋 🍎 🍓 🍒

Emergent visual-language capabilities
Zero-shot multimodal chain of thought CoT, multi-image reasoning



Given ****. Q: Can I go down this street on a bicycle, yes or no? A: Let's think step by step.
1. do not enter. 2. except bicycles. 3. do not entry except bicycles. 4. yes.

zero-shot generalization
(unseen object pairings, or objects)



Visually-conditioned Jokes + Few-shot prompting



I see (image omitted). Description: A cow sitting on a rug. Joke: Why is the cow sitting on a rug? Because it wants to be a cow-ch! ... I see . Description: a donkey is carrying a dog, cat, and rooster. Joke: what do you call a donkey with a rooster on his back. A rooster booster.

Zero-shot Multimodal Chain-of-Thought



Given . Q: Can I go down this street on a bicycle, yes or no? A: Let's think step by step. 1. do not enter. 2. except bicycles. 3. do not entry except bicycles. 4. yes.



Given . Q: How many championship rings did the player on the left win in his career A: Let's think step by step. 1. kobe bryant is on the left. 2. kobe bryant has won 5 championship rings.



Given . Q: Who are the two teams playing in this photo, and which was the last to win a championship, and which year did they win, and who was their star player that year? A: Let's think step by step. 1. team in white is new york knick. 2. team in green is boston celtic. 3. last to win championship was boston celtic. 4. year was 2008. 5. star player was paul pierce. 6. i am not sure if this is correct. i am not basketball fan. i just google it.

Robot Visual Perception, Dialogue, and Planning



Robot: I am a robot operating in a kitchen. Given , what do I see? a bowl of eggs, a bowl of flour, and a bowl of sugar.. and a spatula.. and a spoon.. and a bag of flour.. and a bag of sugar. When a human asks me to do a task, I will respond with the sequence of actions I would do to accomplish the task with only the items I see. Human: Use all of the ingredients you see to make a cake batter. 1. crack egg. 2. put egg in bowl. 3. put flour in bowl. 4. put sugar in bowl. 5. mix. 6. put in pan. 7. bake. 8. eat. 9. clean up.

Physical Prediction



Given . Q: What will the robot do next? A: fall.

Spatial Grounding



Given . Q: What flavor is the doughnut on the right? Strawberry.. Q: What flavor is the doughnut on the left? A blueberry.

Robot Initiative



Given . Q: If a robot wanted to be useful here, what steps should it take? A: clean table..pick up trash.. pick up chairs.. wipe chairs.. put chairs down.

Zero-shot: Multi-image Relationships



Q: Where is in this scene ? A: a stove top with a pot on it.



Photo 1: . Photo 2: . Q: What is in Photo 1 but NOT in Photo 2? A: sunglasses.

Photo 1: . Photo 2: . Q: What is in Photo 1 but NOT in Photo 2? A: Let's think step by step and explain. Photo 1 has sunglasses on top of folded clothes. Photo 2 does not have sunglasses on top of folded clothes. Therefore, sunglasses are in Photo 1 but not in Photo 2.

End-to-End Egocentric Q&A, Dialogue



Photo 1, at 10:30 am: . Photo 2, at 12:45 pm: . Photo 3, at 3:45 pm: . Q: I forget, what did I have for lunch, and what time was it? A: Let's think step by step. 1. you had a sandwich for lunch. 2. it was 12:45 pm.

OCR-free Math Reasoning



Given . Q: I am getting just two custom pizza for me and my friend. How much should I pay in total? A: Let's think step by step. 1st step: 2 custom pizzas. 2nd step: 2 x \$9.99. 3rd step: \$19.98.

Large Language Models

- demonstrate **strong reasoning capabilities** across various domains:
 - dialogue
 - step-by-step reasoning
 - math problem solving
 - code writing
- **grounding** issue:
 - Connecting the **LLM representations** to real-world visual and physical sensor modalities
 - is essential to solve **grounded real-world problems**
 - where the **geometric configuration** of the scene is important:
 - sensory experiences such as vision
 - actions or interactions in the physical world
 - anchors in factual, structured knowledge bases
 - cultural nuances, situational contexts, or social dynamics

Embodied Language Model

- directly incorporates continuous inputs from **sensor modalities**
- conducts more **grounded** decision making in the real world
- Input
 - **images**
 - **state estimates**
 - is embedded into the same latent space as **language tokens** and
 - is processed by the **self-attention layers** of a transformer-based LLM
 - in the same way as **text**.

Research Results

- **Multi-task** training improves performance of the trained embodied language model.
- **Transfer across tasks** leads to high data efficiency for
 - robotics tasks
 - zero-shot multimodal chain-of-thought (CoT) reasoning
 - few-shot prompting
 - OCR-free math reasoning
 - multi-image reasoning
- A generalist, transfer-learned, multi-embodied decision-making **agent**
- **can be trained**
- via mixing **embodied data into** the training of a (then) multimodal **LLM**.

Main Architectural Idea

- Inject **continuous embodied observations**
 - images
 - state estimates
 - sensor modalities
- into language embedding space of a **pretrained model**.
 - PaLM as pretrained model
 - Chowdhery, A., Narang, S., Devlin, J., Bosma, M., Mishra, G., Roberts, A., ... & Fiedel, N. (2022). Palm: Scaling language modeling with pathways. arXiv preprint arXiv:2204.02311.
- Encode **continuous observations** into a sequence of vectors with the same dimension as the embedding space of the language tokens.
- Input: multi-modal sentence
 - Q: What happened between <img_1> and <img_2>?

Decoder-Only LLM

- generative model trained to predict probability p of a piece of text
- that is represented by a sequence of tokens w

$$p(w_{1:L}) = \prod_{l=1}^L p_{\text{LM}}(w_l | w_{1:l-1})$$

- prefix-decoder-only LLM

$$p(w_{n+1:L} | w_{1:n}) = \prod_{l=n+1}^L p_{\text{LM}}(w_l | w_{1:l-1})$$

- Prefix provides the context.
 - e.g. description of the task the LLM should solve
- The LLM predicts the continuation of the prefix

Input and Scene Representations for Different Sensor Modalities

- Modalities and encoders
 - State estimation vector $\mathbf{s}$
 - MLP maps $\mathbf{s}$ into embedding space
 - Vision transformer (ViT)
 - mapping image I into a number of token embeddings
 - token: see chapter on transformer
 - Object-centric representations
 - separate visual inputs into distinct objects before injecting them into LLM
 - conceptually requiring ground truth object instance masks
 - Object Scene Representation Transformer (OSRT)
 - 3D scene neural representation
 - providing several object slots
 - each projected with an MLP

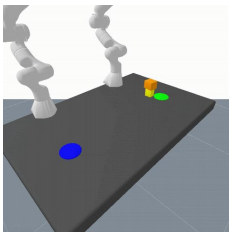
Experiments

in robotics

- Task and Motion Planning (TAMP)
 - multi-object tabletop pushing environment
 - from Language-Table dataset
 - publicly available
- Mobile Manipulation Environment

but also

- General Visual-Language Task
- General Language Task



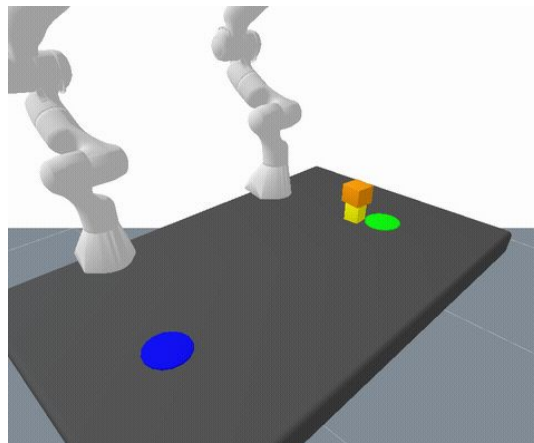
learn

encoder
 ϕ

Multimodal
LLM

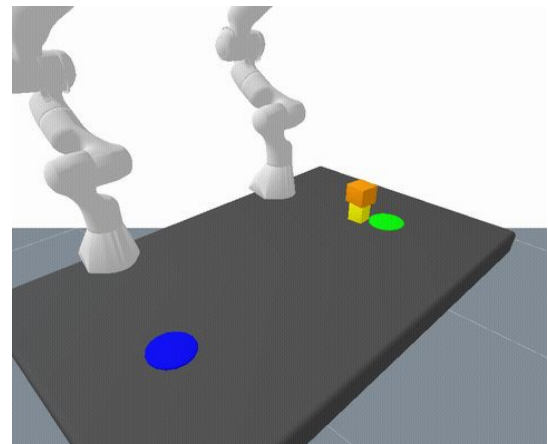
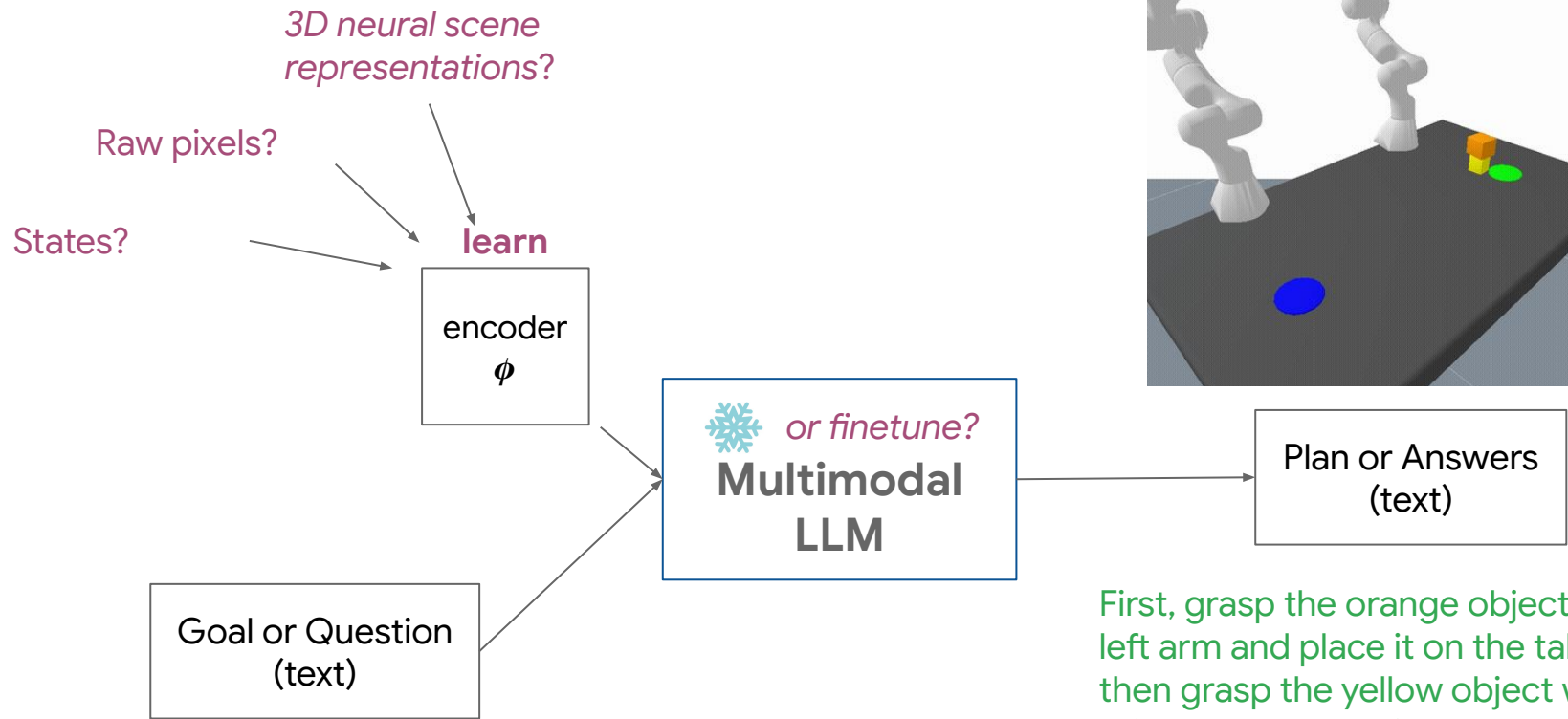
Goal or Question
(text)

How to move the yellow block
on top of the blue plate?



Plan or Answers
(text)

First, grasp the orange object with the left arm and place it on the table, then grasp the yellow object with the left arm and place it on the table close enough to the right arm, then grasp the yellow object with the right arm and place it on the blue plate.



How to move the yellow block
on top of the blue plate?

First, grasp the orange object with the left arm and place it on the table, then grasp the yellow object with the left arm and place it on the table close enough to the right arm, then grasp the yellow object with the right arm and place it on the blue plate.

Research Questions

1

Can **multimodal features** (pixels or states) be input to an LLM while leveraging the **generalization of LLMs** for embodied reasoning and planning?

2

How to form multimodal input representations for training an embodied LLM?

- Inputs: *states, pixels (ViT), neural 3D scenes*
- Entity-based interleaving (object-centric)

3

Other **key aspects** of training these models?

- LLM: pretrained? frozen? finetuned?
- Encoder: pretraining?
- Data: sample complexity?

Research Questions

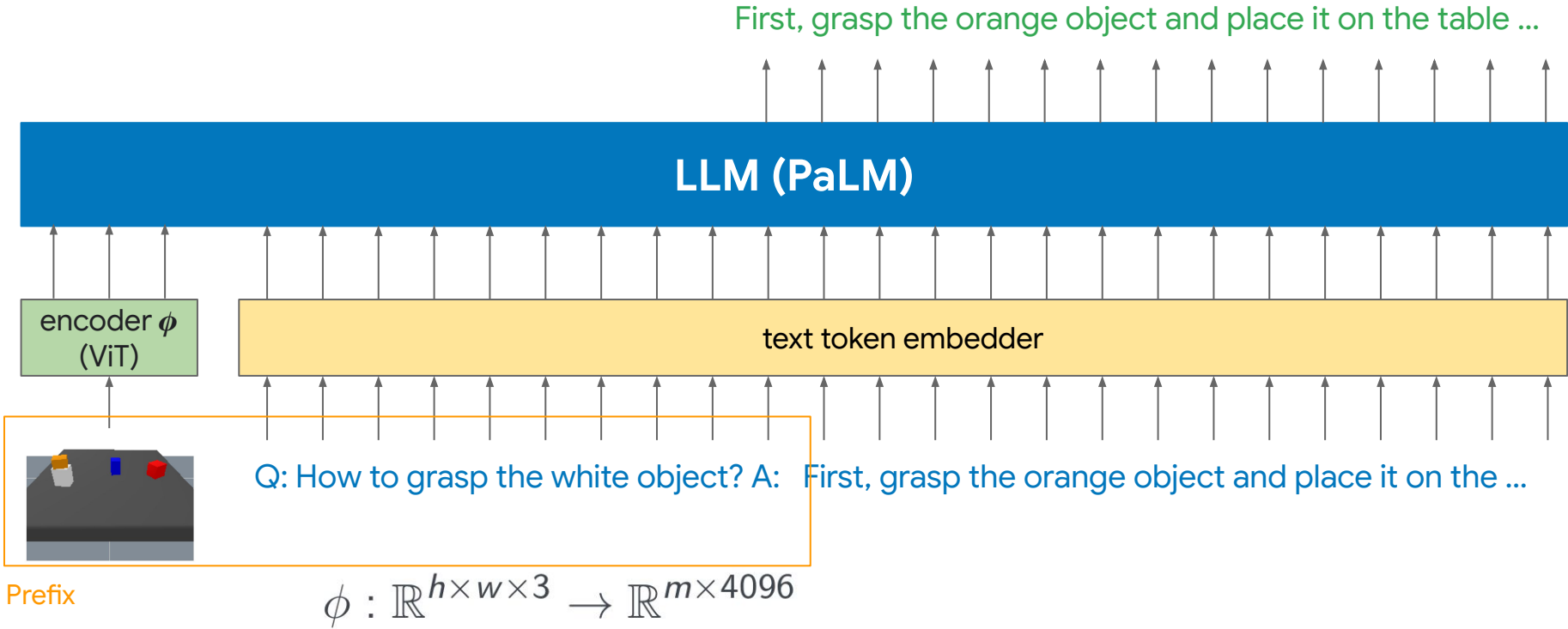
4

Large-scale model + large scale data – do we get *transfer*? What *else*?

- Scaling the LLM (8B, 62B, 562B, ...)
- Scaling the ViT (4B, 22B, ...)
- Scaling to all robotics data
- Co-training on general VQA tasks

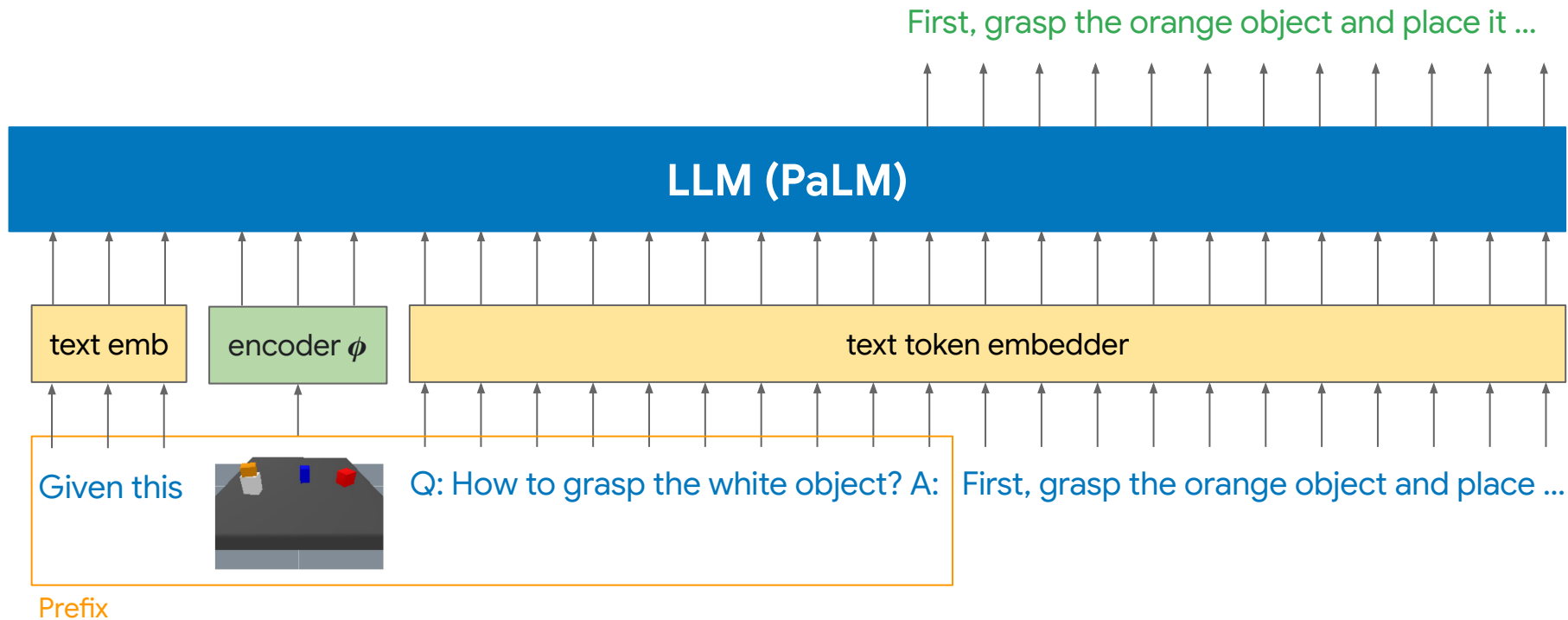
- Minimizing catastrophic forgetting
- Emergent capabilities: zero-shot multimodal Chain of Thought (CoT), multi-image reasoning
- The model, **given only the prompt**, will spell out a **step-by-step reasoning** narrative that **mixes textual and visual** (or other) evidence before stating its final answer.

PaLM-E – Injecting multimodal information into PaLM



Prefix

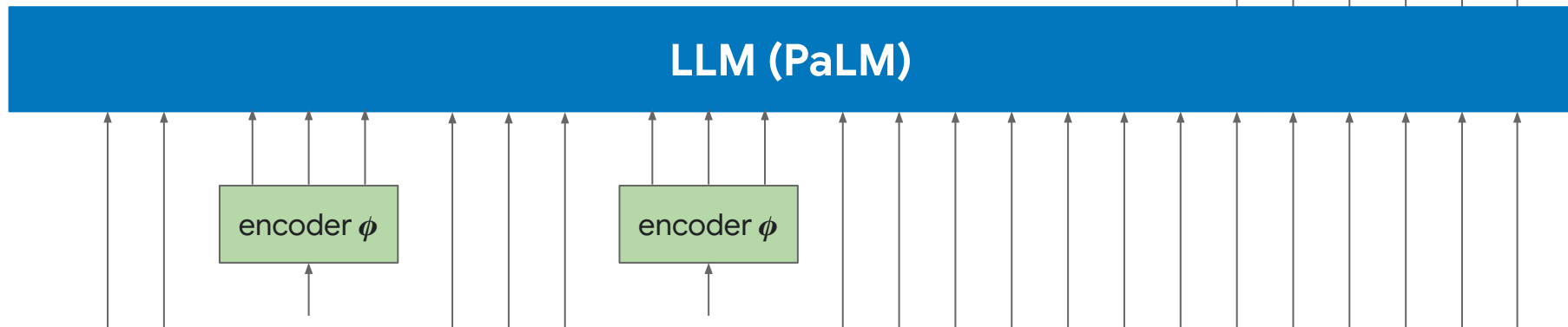
PaLM-E – Injecting multimodal information into PaLM



Entity-based Interleaving of Text and Continuous Inputs

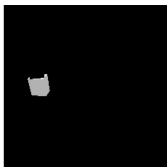
enable LLM to refer to entities!

First, grasp object 2, then



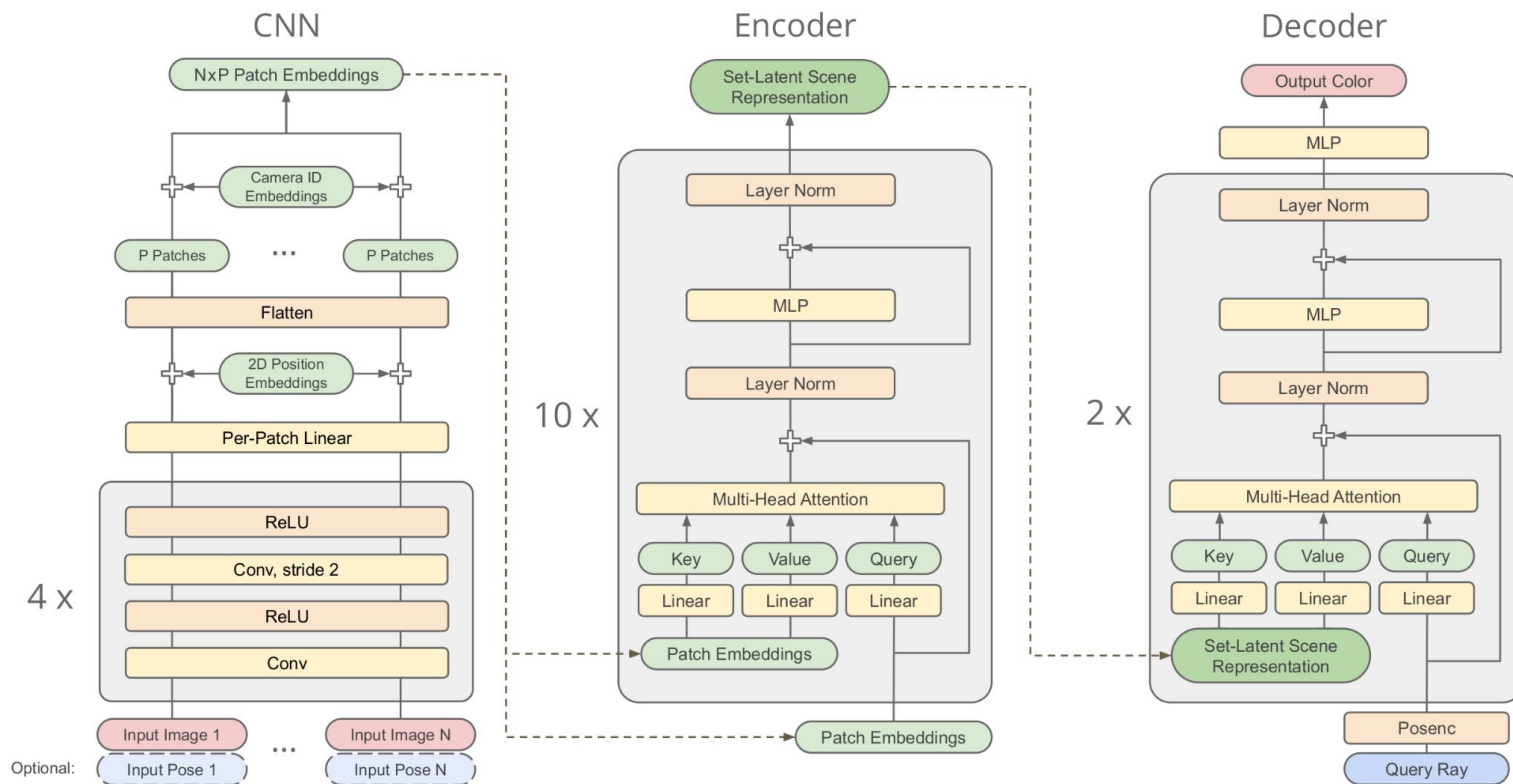
Object 1 is <multim_tok_1>. Object 2 is <multim_tok_2>. Q: How to grasp object 1?

A: First, grasp object 2



Multimodal
Prefix

Scene Representation Transformers



Scene Representation Transformers

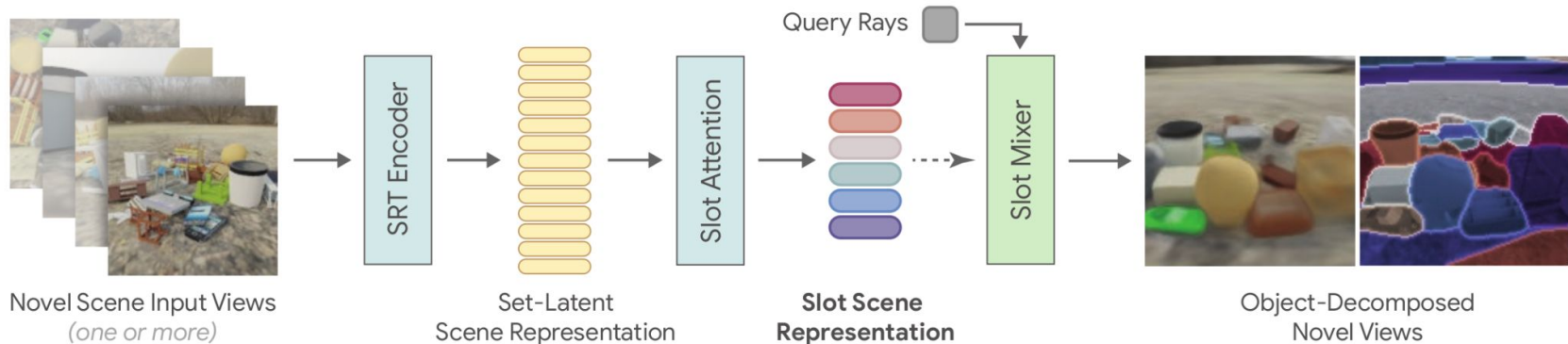
- **Geometry-free** novel view synthesis
- One data point used for training is
a set s of RGB input images $\{I_i \in \mathbb{R}^{H \times W \times 3}\}$
from the scene. (Training is done with a lot of those sets!)
- CNN extracts features which are flattened and combined into a single set of tokens.
- Encoder Transformer $\mathcal{E}$ performs **self-attention** yielding the
Set-Latent Scene Representation (SLSR)
 $\{\mathbf{z}_j \in \mathbb{R}^d\} = \mathcal{E}_\theta(\{\text{CNN}_\theta(\mathbf{I}_i)\})$
- Decoder Transformer $\mathcal{D}$ uses ray $\mathbf{r}$ to **attend** into SLSR to predict RGB image

$$C(\mathbf{r}) = \mathcal{D}_\theta(\mathbf{r} \mid \{\mathbf{z}_j\}) \quad \mathbf{r} = (\mathbf{o}, \mathbf{d}) = (\text{camera}, \text{ray direction})$$

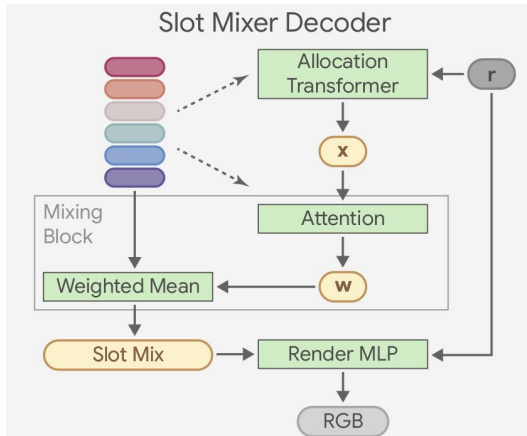
- Model is trained end to end: $\arg \min_{\theta} \sum_s \mathbb{E}_{\mathbf{r} \sim \mathbf{I}_{s,i}^{\text{gt}}} \|C(\mathbf{r}) - \mathbf{I}_{s,i}^{\text{gt}}(\mathbf{r})\|_2^2$

Object Scene Representation Transformers

- Slot Attention module converts SLSR into object-centric Slot Scene Representation

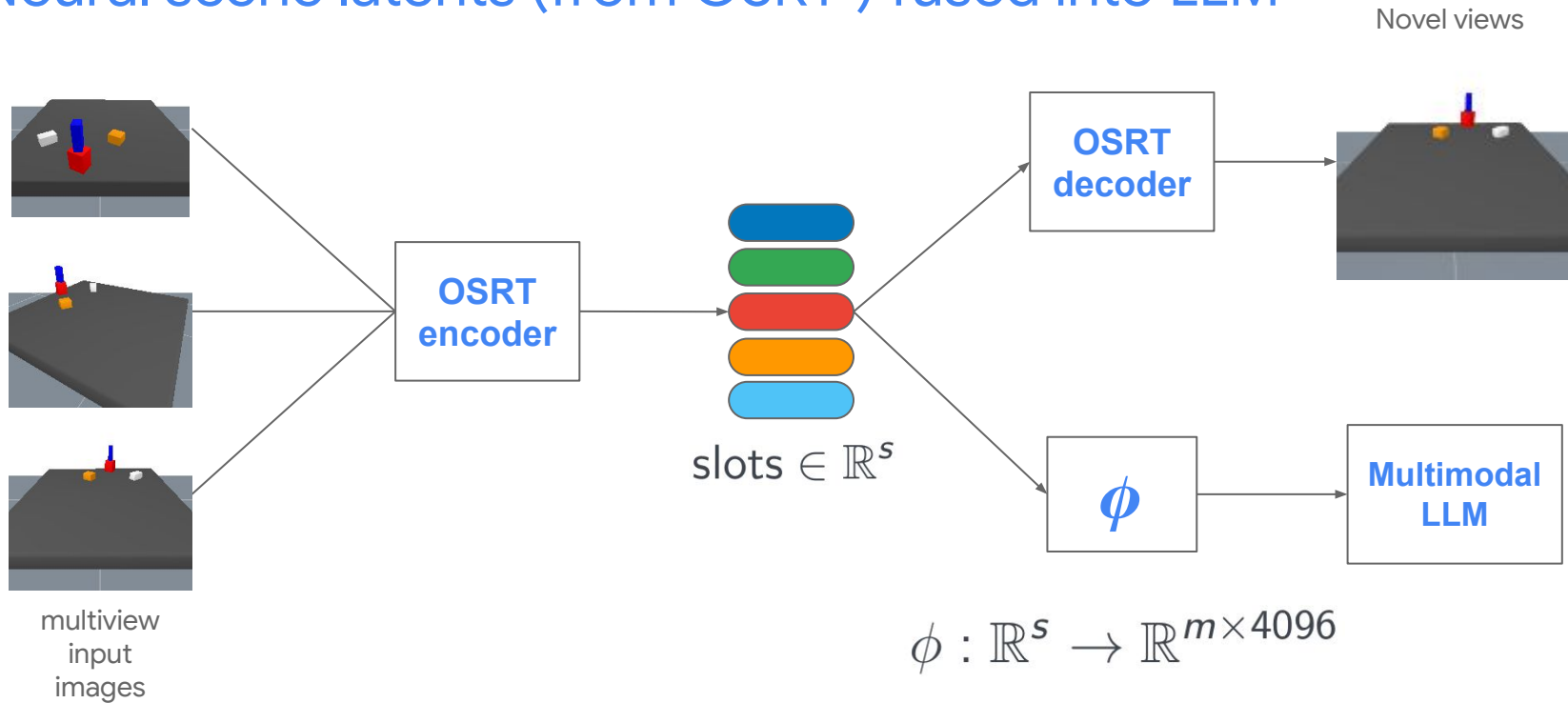


- SlotSR has one-to-one correspondence between tokens and objects.
- Slot Mixer is used to decode SlotSR to object-decomposed views.
 - Allocation Transformer queries SlotSR using ray
 - Attention layer for mixing
 - Rendering MLP for decoding



Sajjadi, M. S., Duckworth, D., Mahendran, A., van Steenkiste, S., ... & Kipf, T. (2022). Object scene representation transformer. Adv. in Neural Information Processing Systems, 35, 9512-9524.

Neural scene latents (from OSRT¹) fused into LLM

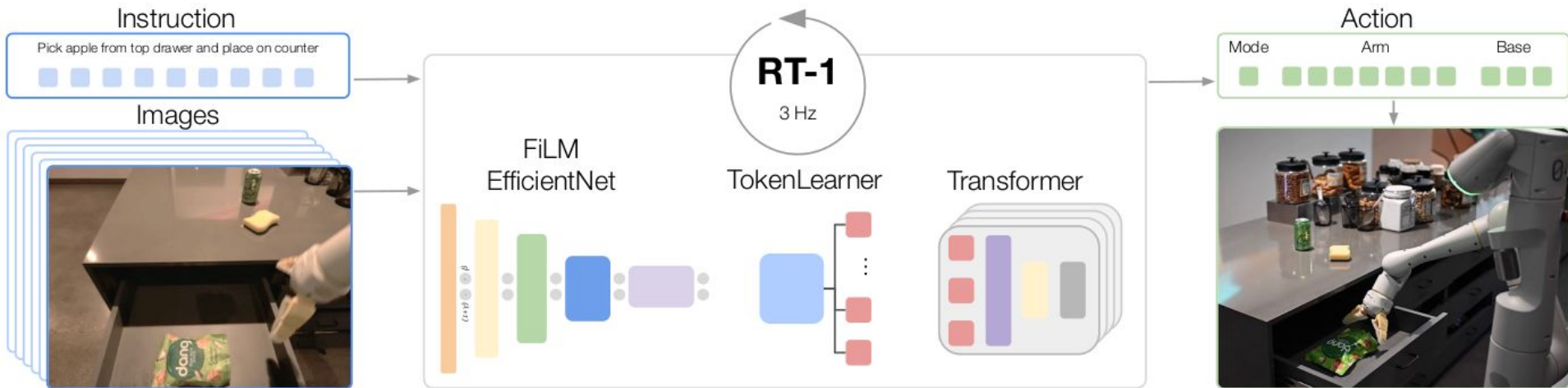


Output of PaLM-E in Robot Control Loop

- Output of PaLM-E is **text generated autoregressively**: a **sequence of decisions** that should be executed by a robot.
- PaLM-E can be understood as a high-level (abstract) policy that conditions the low-level policies of a robot.
- The **high-level** output text **sequences and conditions** the **low-level** commands.
- PaLM-E must determine **which skills are available** based on training data and prompt.
- PaLM-E is integrated into a control loop.
- A low-level policy or **planner** **translates decisions into a sequence of low-level actions**
- The planner provides a solution to the **embodied planning** or **control task**.
- Executed **actions lead to new observations**.

RT-1 Robotics Transformer for Control

- RT-1 takes **images and natural language instructions** and outputs **discretized base and arm actions**.



Brohan, A., Brown, N., Carbajal, J., Chebotar, Y., Dabis, J., Finn, C., ... & Zitkovich, B. (2022). Rt-1: Robotics transformer for real-world control at scale. *arXiv preprint arXiv:2212.06817*.

Zitkovich, B., Yu, T., Xu, S., Xu, P., Xiao, T., Xia, F., ... & Han, K. (2023, December). Rt-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning* (pp. 2165-2183). PMLR.

RT-2 Vision-Language-Action Model

Internet-Scale VQA + Robot Action Data



Q: What is happening in the image?

A: 311 423 170 55 244

A grey donkey walks down the street.

Q: Que puis-je faire avec ces objets?

A: 3455 1144 189 25673

Faire cuire un gâteau.



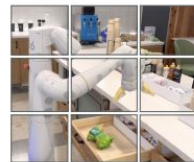
Q: What should the robot do to **<task>**?

A: 132 114 128 5 25 156

$\Delta T = [0.1, -0.2, 0]$
 $\Delta R = [10^\circ, 25^\circ, -7^\circ]$

Vision-Language-Action Models for Robot Control

Q: What should the robot do to **<task>**? A: ...



RT-2

ViT

Large Language Model

A: 132 114 128 5 25 156

De-Tokenize

$\Delta T = [0.1, -0.2, 0]$
 $\Delta R = [10^\circ, 25^\circ, -7^\circ]$

Robot Action

Co-Fine-Tune

Deploy

Closed-Loop Robot Control



Put the strawberry into the correct bowl



Pick the nearly falling bag

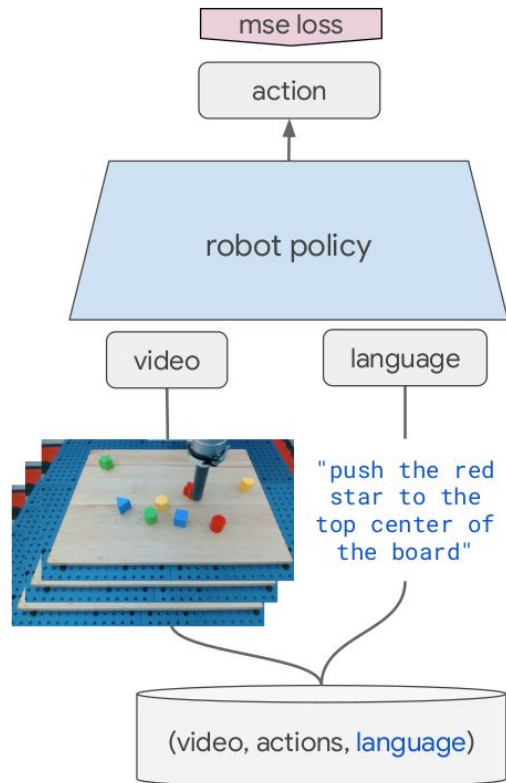


Pick object that is different

Brohan, A., Brown, N., Carbajal, J., Chebotar, Y., Dabis, J., Finn, C., ... & Zitkovich, B. (2022). Rt-1: Robotics transformer for real-world control at scale. *arXiv preprint arXiv:2212.06817*.

Zitkovich, B., Yu, T., Xu, S., Xu, P., Xiao, T., Xia, F., ... & Han, K. (2023, December). Rt-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning* (pp. 2165-2183). PMLR.

Language-Conditioned Behavior



Results

Embodied VQA and Planning

	Object- centric	LLM pre-train	Embodied VQA				Planning	
			q ₁	q ₂	q ₃	q ₄	p ₁	p ₂
SayCan (oracle afford.) (Ahn et al., 2022)		✓	-	-	-	-	38.7	33.3
PaLI (zero-shot) (Chen et al., 2022)		✓	-	0.0	0.0	-	-	-
<i>PaLM-E</i> (ours) w/ input enc:								
State	✓(GT)	✗	99.4	89.8	90.3	88.3	45.0	46.1
State	✓(GT)	✓	100.0	96.3	95.1	93.1	55.9	49.7
ViT + TL	✓(GT)	✓	34.7	54.6	74.6	91.6	24.0	14.7
ViT-4B single robot	✗	✓	-	45.9	78.4	92.2	30.6	32.9
ViT-4B full mixture	✗	✓	-	70.7	93.4	92.1	74.1	74.6
OSRT (no VQA)	✓	✓	-	-	-	-	71.9	75.1
OSRT	✓	✓	99.7	98.2	100.0	93.7	82.5	76.2

Zero-shot Baselines						Task 1			Task 2			Task 3		
SayCan (oracle afford.) (Ahn et al., 2022)						0.0			-			-		
PaLI (Chen et al., 2022)						0.0			-			-		
PaLM-E-	trained on	from scratch	LLM+ViT pretrain	LLM frozen	Task finetune	# Demos		40	10	20	40	10	20	80
12B	Single robot	✓	✗	n/a	✓	20.0	30.0	50.0	2.5	6.3	2.5	11.3	16.9	28.3
12B	Full mixture	✗	✓	✓	✗	-	-	20.0	-	-	36.3	-	-	29.4
12B	Full mixture	✗	✓	✗	✗	-	-	80.0	-	-	57.5	-	-	50.0
12B	Full mixture	✗	✓	✗	✓	70.0	80.0	80.0	31.3	58.8	58.8	57.5	54.4	56.3
84B	Full mixture	✗	✓	✗	✗	-	-	90.0	-	-	53.8	-	-	64.4

Task 1. Q: There is a block that is closest to $\{i.e., \text{top right corner}\}$. Push that block to the other block of the same color.

Task 2. Q: How to sort the blocks by colors into corners?

Task 3. Q: How to push all the blocks that are on the $\{\text{left/right}\}$ side together, without bringing over any of the blocks that are on the $\{\text{right/left}\}$ side?

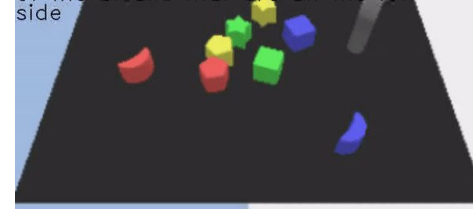
instruction: push the red pentagon to the top left corner, reward: -2.78, lh_instruction: sort the blocks by colors into different corners



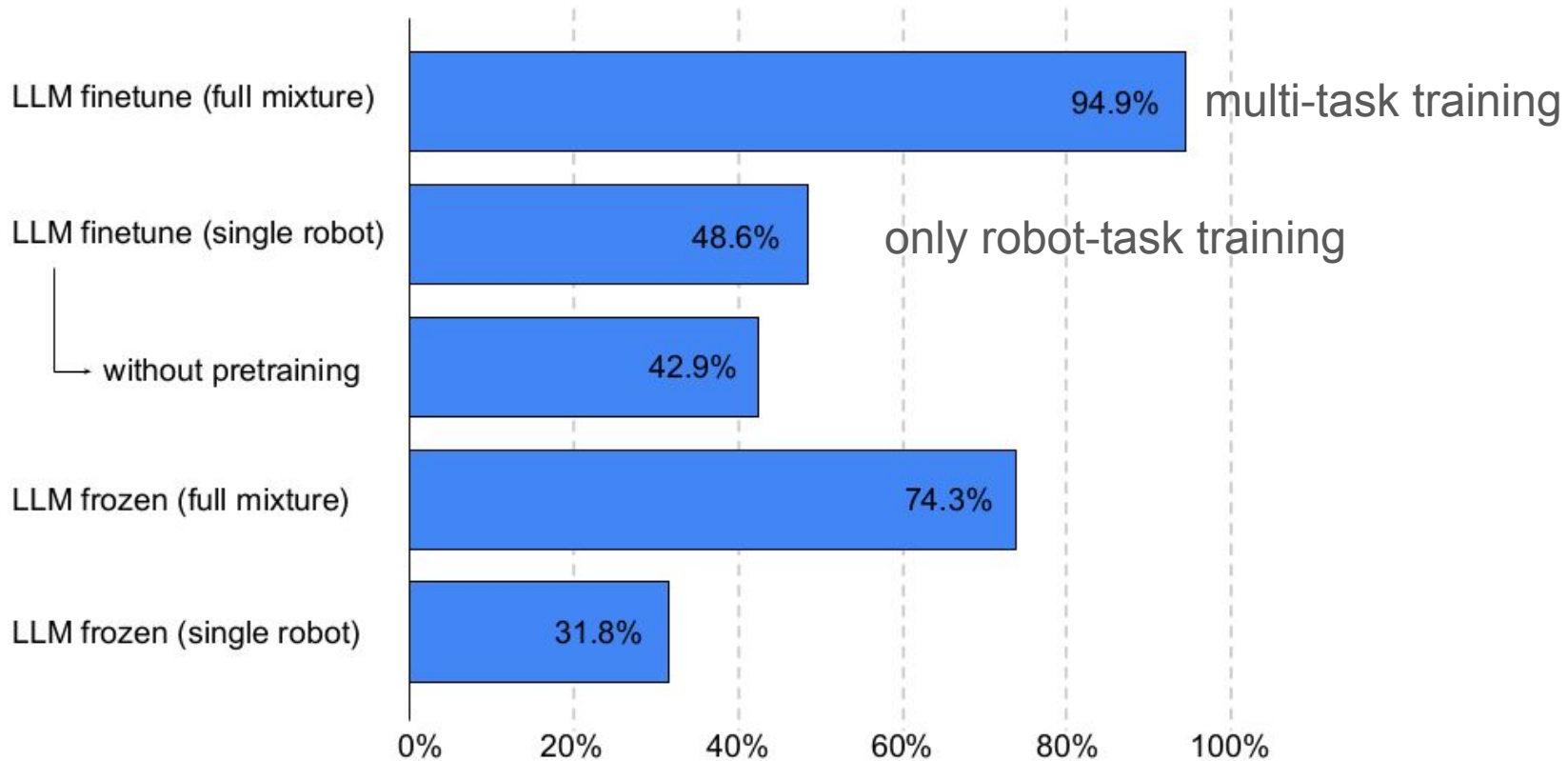
instruction: push the blue moon to the yellow pentagon, reward: 0.37, lh_instruction: push all the blocks that are on the left side together, without bringing over any of the blocks that are on the right side



instruction: push the blue moon to the blue cube, reward: 0.42, lh_instruction: push all the blocks that are on the right side together, without bringing over any of the blocks that are on the left side



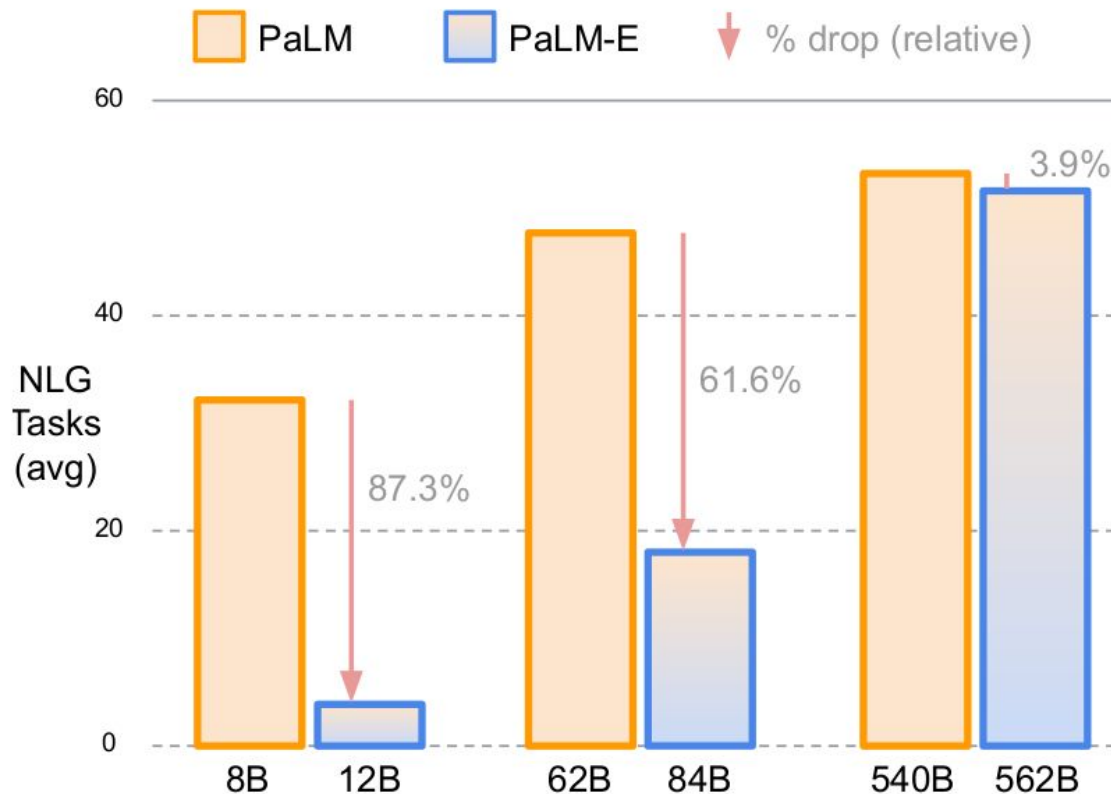
Planning Success Results



PaLM-E 562B is also a visual-language generalist

Model	VQAv2		OK-VQA	COCO
	test-dev	test-std	val	Karpathy test
<i>Generalist (one model)</i>				
PaLM-E-12B	76.2	-	55.5	135.0
PaLM-E-562B	80.0	-	66.1	138.7
<i>Task-specific finetuned models</i>				
Flamingo (Alayrac et al., 2022)	82.0	82.1	57.8†	138.1
PaLI (Chen et al., 2022)	84.3	84.3	64.5	149.1
PaLM-E-12B	77.7	77.9	60.1	136.0
PaLM-E-66B	-	-	62.9	-
PaLM-E-84B	80.5	-	63.3	138.0
<i>Generalist (one model), with frozen LLM</i>				
(Tsimpoukelli et al., 2021)	48.4	-	-	-
PaLM-E-12B frozen	70.3	-	51.5	128.0

Catastrophic forgetting reduced with scale



Conclusions

- Embodied reasoning as vision-language modeling
- Transfer across **robot** and general **vision-language** datasets
- PaLM-E also competitive in general VQA, captioning, language tasks
- Other input representations (OSRT, state)

PaLM: Pathways Language Model

- Chowdhery, A., Narang, S., Devlin, J., Bosma, M., Mishra, G., Roberts, A., ... & Fiedel, N. (2022). Palm: Scaling language modeling with pathways. arXiv preprint arXiv:2204.02311.

PaLI: Pathways Language and Image Model

- Chen, X., Wang, X., Changpinyo, S., Piergiovanni, A. J., Padlewski, P., Salz, D., ... & Soricut, R. (2022). Pali: A jointly-scaled multilingual language-image model. arXiv preprint arXiv:2209.06794.

OSRT: Object Scene Representation Transformer

- Sajjadi, M. S., Duckworth, D., Mahendran, A., van Steenkiste, S., Pavetic, F., Lucic, M., ... & Kipf, T. (2022). Object scene representation transformer. Advances in Neural Information Processing Systems, 35, 9512-9524.

SayCan Algorithm

- Ahn, M., Brohan, A., Brown, N., Chebotar, Y., Cortes, O., David, B., ... & Zeng, A. (2022). Do as i can, not as i say: Grounding language in robotic affordances. arXiv preprint arXiv:2204.01691.

Vision transformer up-scaling

- Dehghani, M., Djolonga, J., Mustafa, B., Padlewski, P., Heek, J., Gilmer, J., ... & Houlsby, N. (2023, July). Scaling vision transformers to 22 billion parameters. In International Conference on Machine Learning (pp. 7480-7512). PMLR.

Multi-object pushing environment

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